Examiner only

Answer **all** *questions.*

1. Students are making the fertiliser potassium nitrate in the laboratory. Most fertilisers are formed by the reaction of an acid and an alkali.

(a) Complete the word equation for the formation of potassium nitrate. [2]

.... + potassium hydroxide $\longrightarrow$ potassium nitrate + ...

(b) To make the fertiliser, it is important that the students are able to exactly neutralise the acid with alkali. To do this the students carefully measured 25 cm^3 of acid into a flask and titrated just enough alkali to neutralise the acid.

The results of the experiment are shown below.

Trial	Volume of alkali added at endpoint (titre) (cm^3)
1	20.5
2	20.8
3	18.5
4	20.0
5	19.9

(i) The reading for trial 3 was considered to be an anomalous result. Give a reason why. [1]

....

(ii) Calculate the mean titre. [1]

Mean titre = ... cm^3

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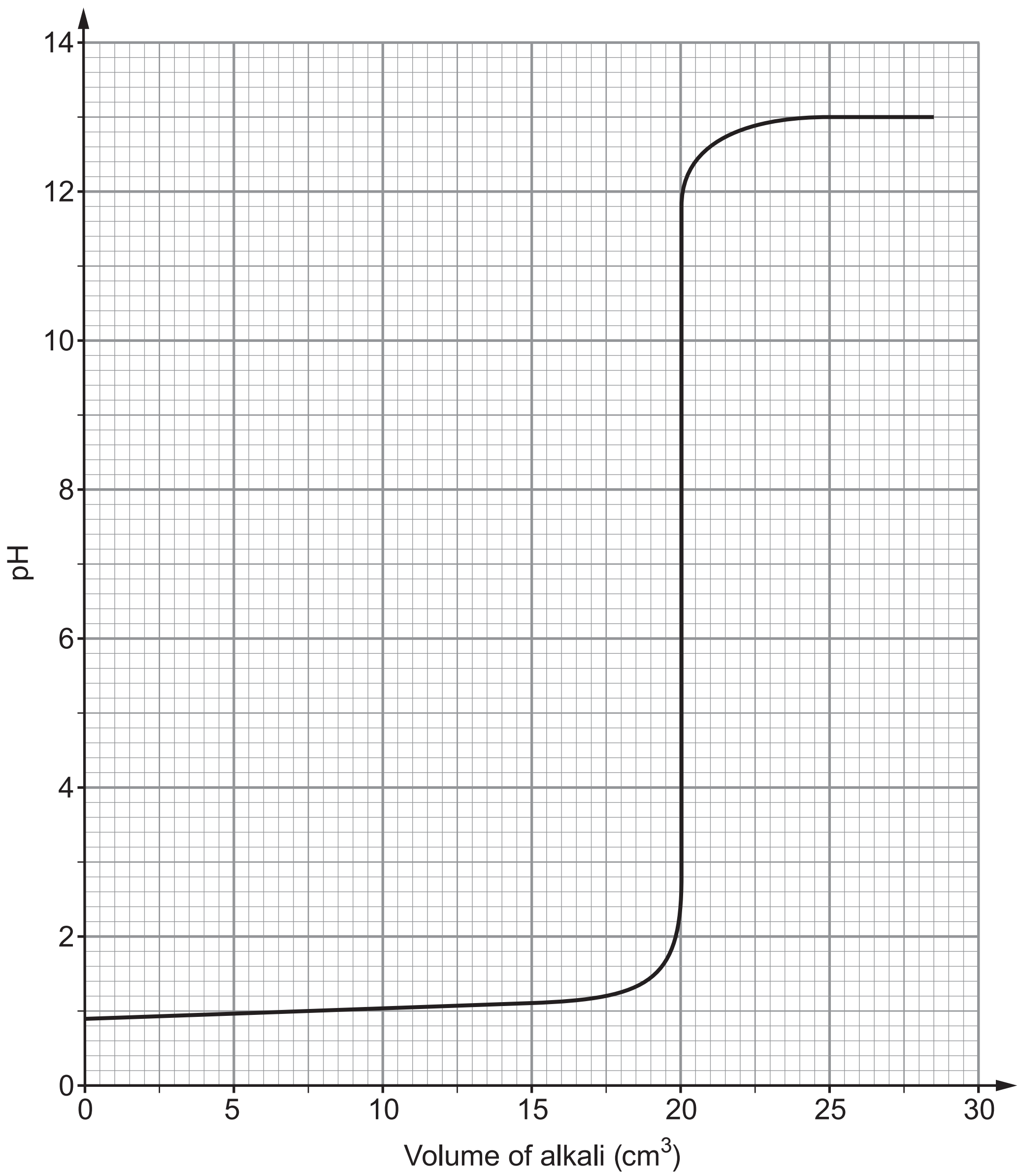
(c) The graph below shows the pH changes, measured by a pH probe, that occurred during trial 4 of the experiment.



Explain the pH changes shown on the graph. [3]

....

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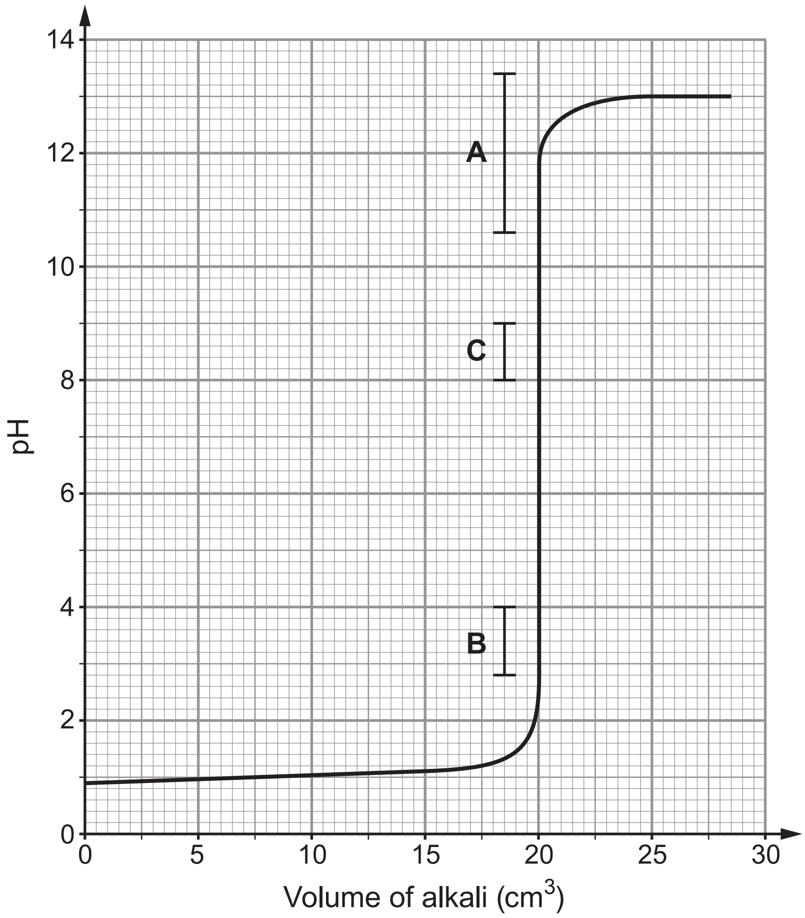
(d) Not all the students had a pH probe. Some had to use an indicator. The pH range of three different indicators is marked on the graph as **A**, **B**, and **C**.



(i) Use the information in the table below to match the range to each indicator. [1]

Indicator	pH range	Colour change	Range on the graph (**A**, **B**, or **C**?)
methyl blue	10.6 - 13.4	blue to violet	....
phenolphthalein	8.0 - 9.0	colourless to pink	....
dinitrophenol	2.8 - 4.0	colourless to yellow	....

(ii) Explain which indicator is not suitable for this experiment. [2]

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